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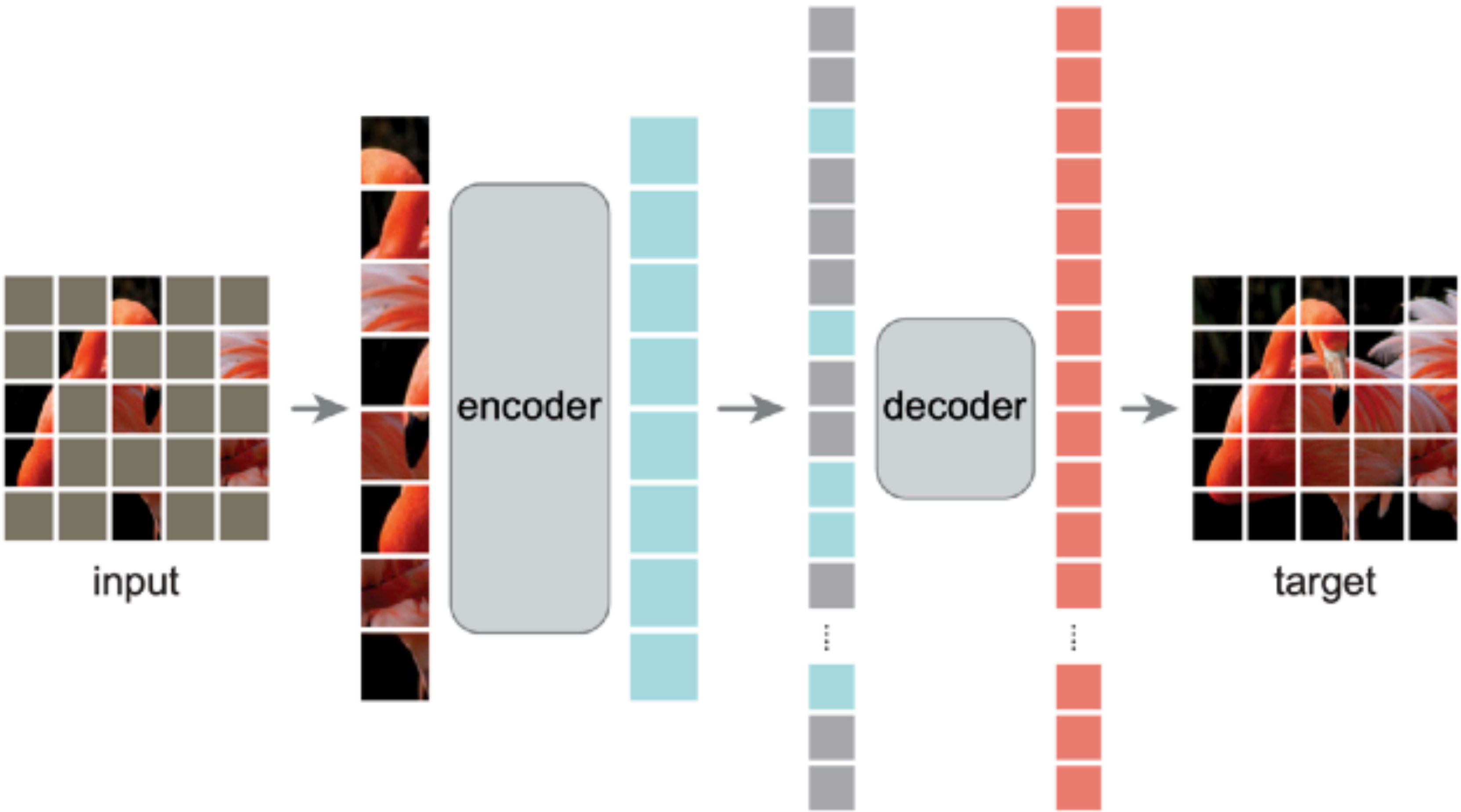
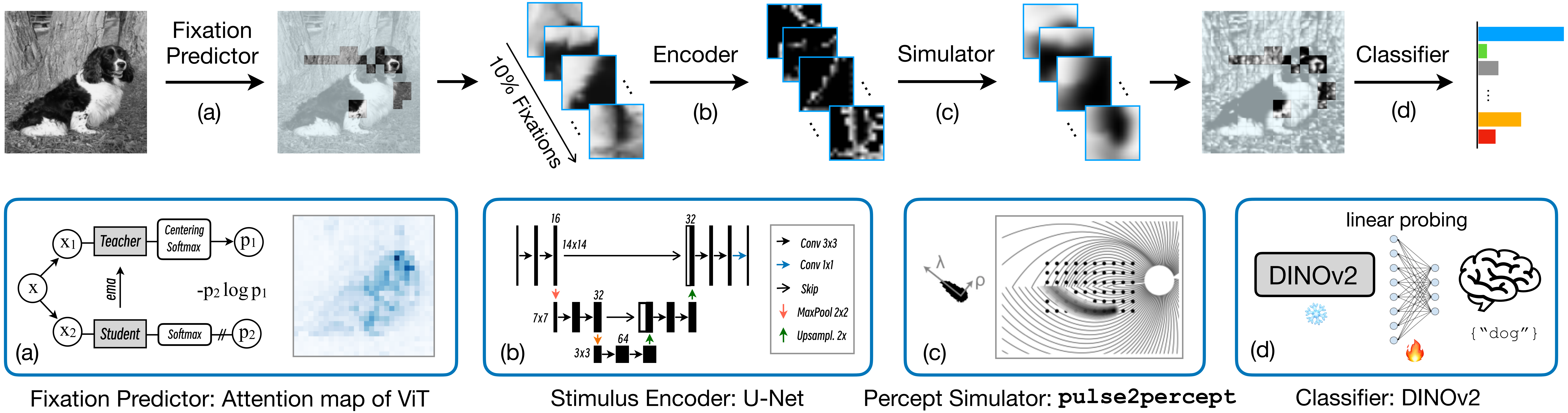
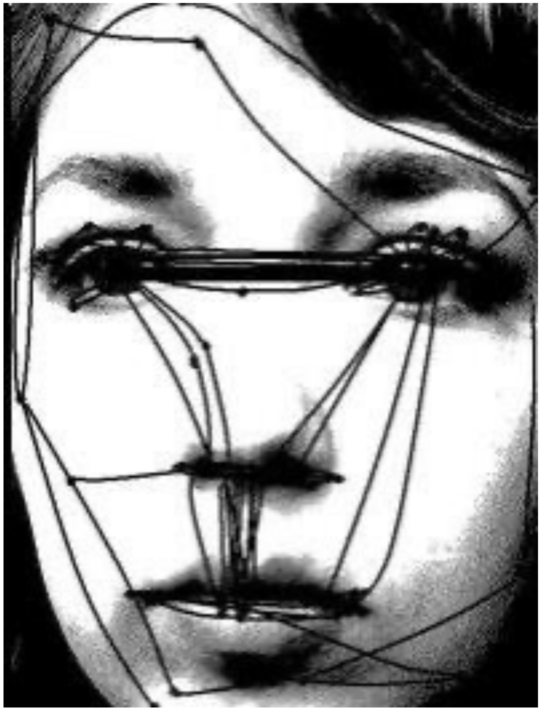

encoder
decoder
input
target

31

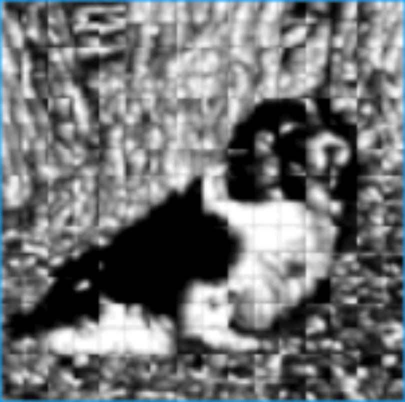
Visual Fixation-based Simulation

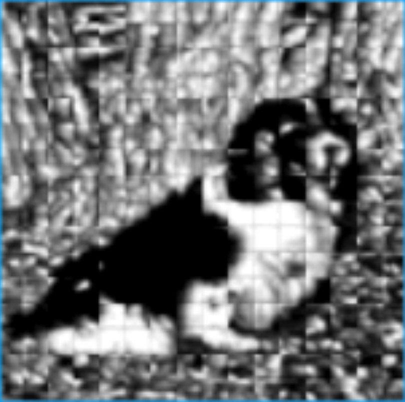

Fixation Predictor
(a)
10% Fixations
Encoder
(b)
Simulator
(c)
Classifier
(d)
Teacher
Centering Softmax
Student
Softmax
ema
x
x_1
x_2
p_1
p_2
$-p_2 \log p_1$
(a)
16
14x14
32
7x7
32
64
3x3
Conv 3x3
Conv 1x1
Skip
MaxPool 2x2
Upsampl. 2x
(b)
λ
ρ
(c)
linear probing
DINOv2
{"dog"}
(d)
Fixation Predictor: Attention map of ViT
Stimulus Encoder: U-Net
Percept Simulator: pulse2percept
Classifier: DINOv2

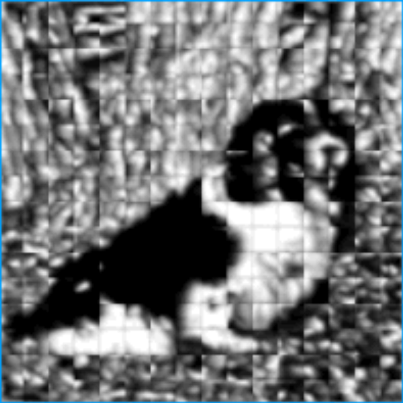


[Wu et al., 2025a]

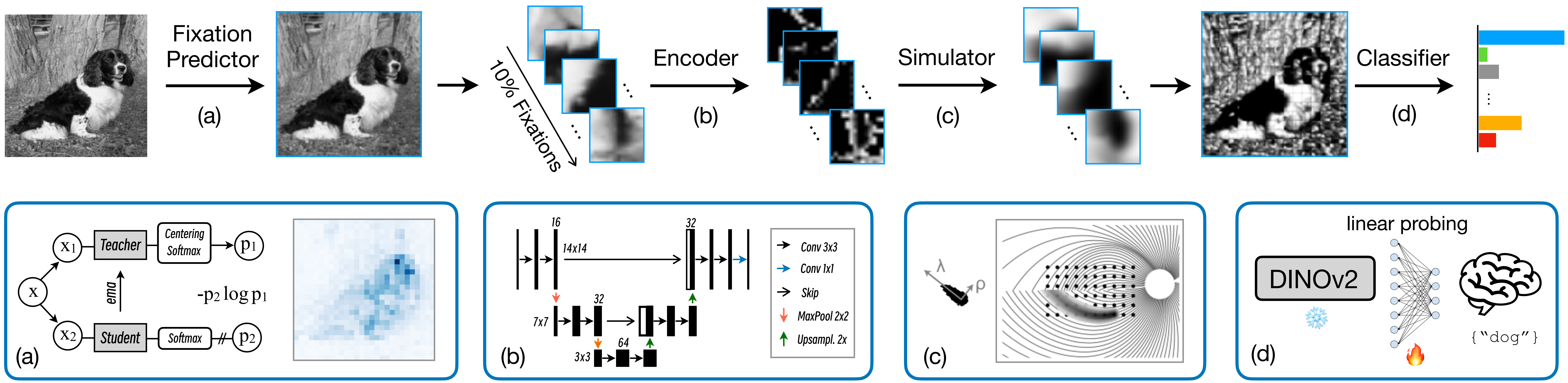
[He et al., 2021]







Visual Fixation-based Simulation

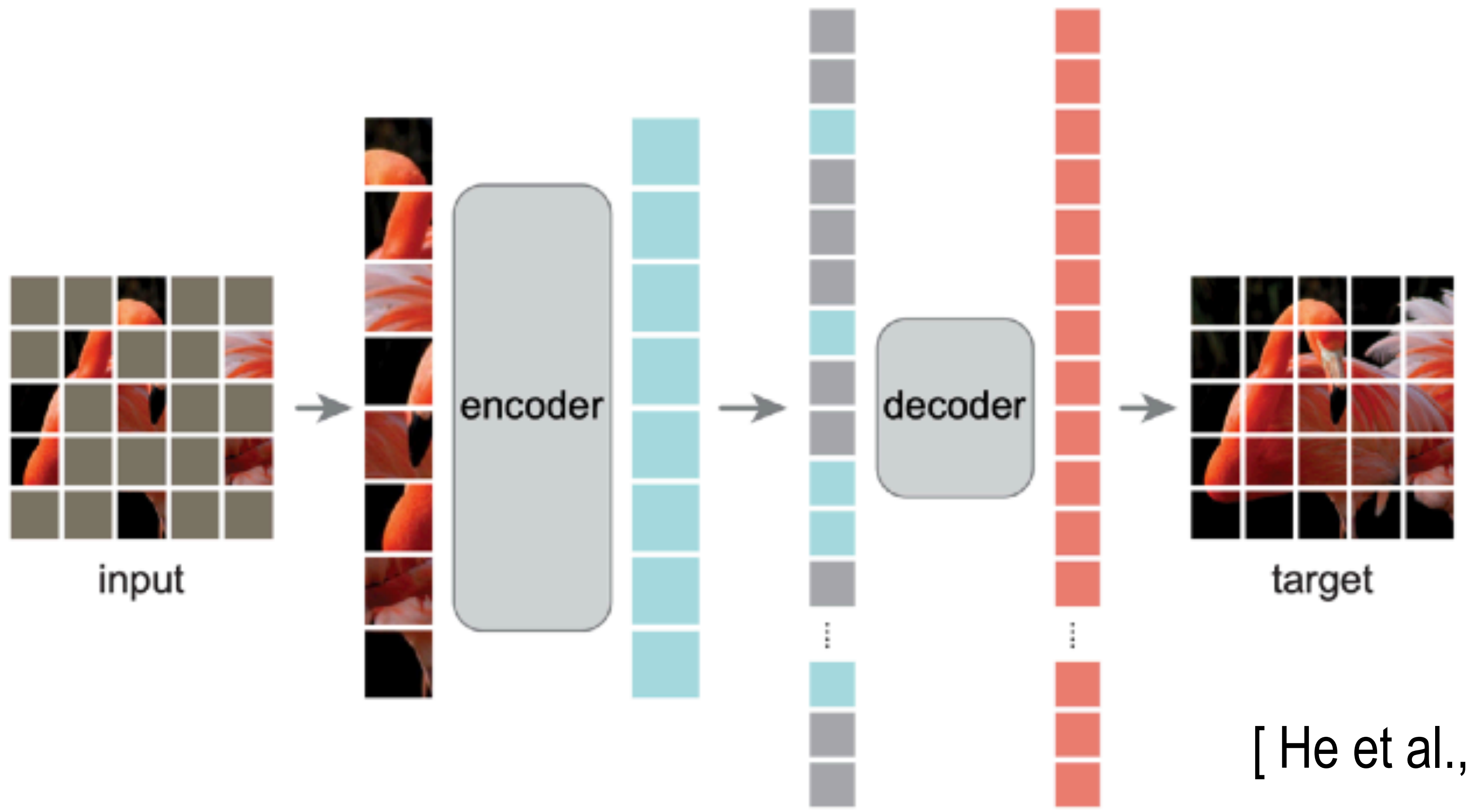


Fixation Predictor: Attention map of ViT

Stimulus Encoder: U-Net

Percept Simulator: `pulse2percept`

Classifier: DINOv2

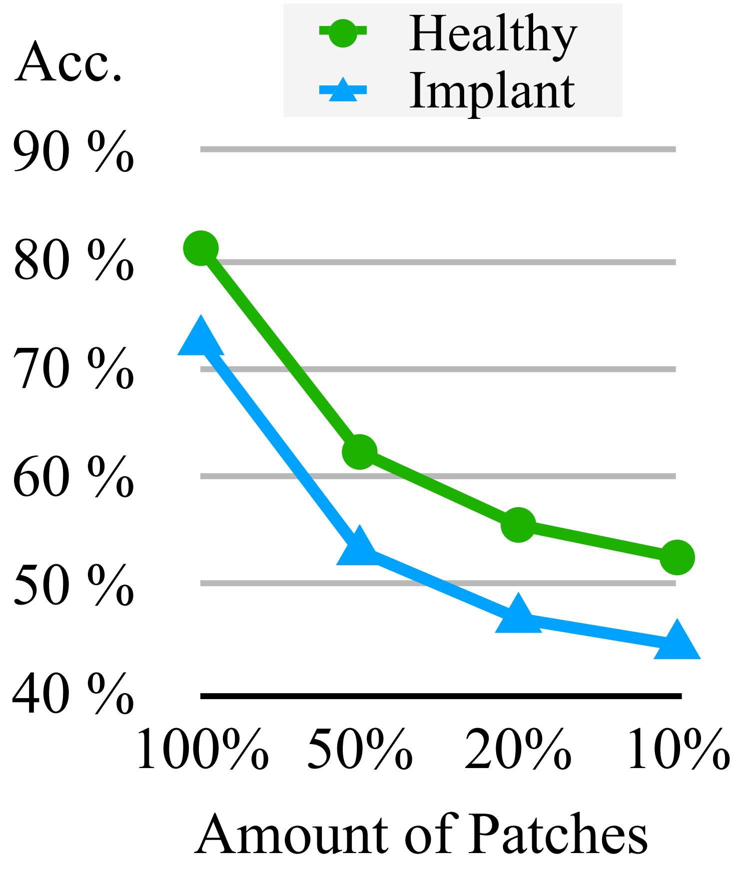


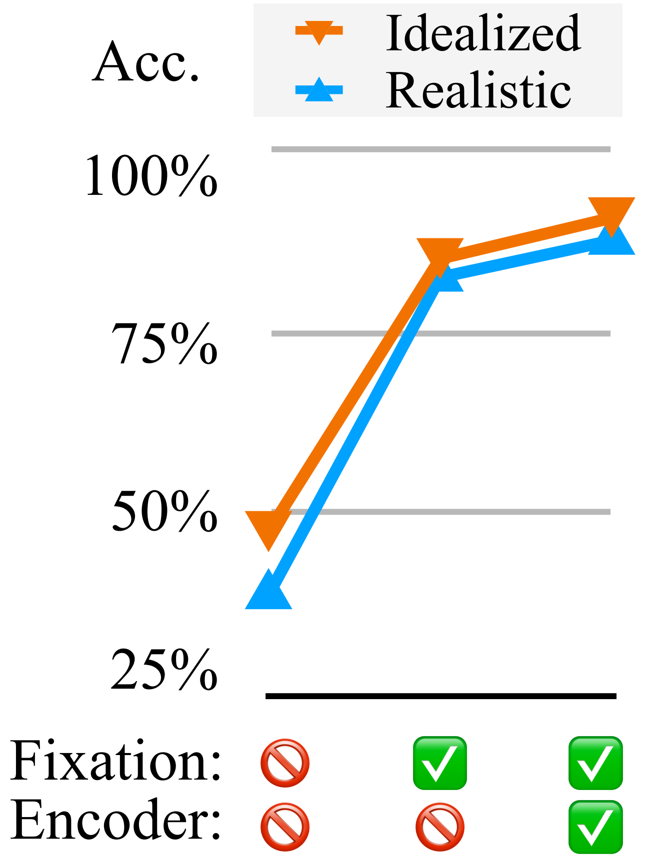
[He et al., 2021]

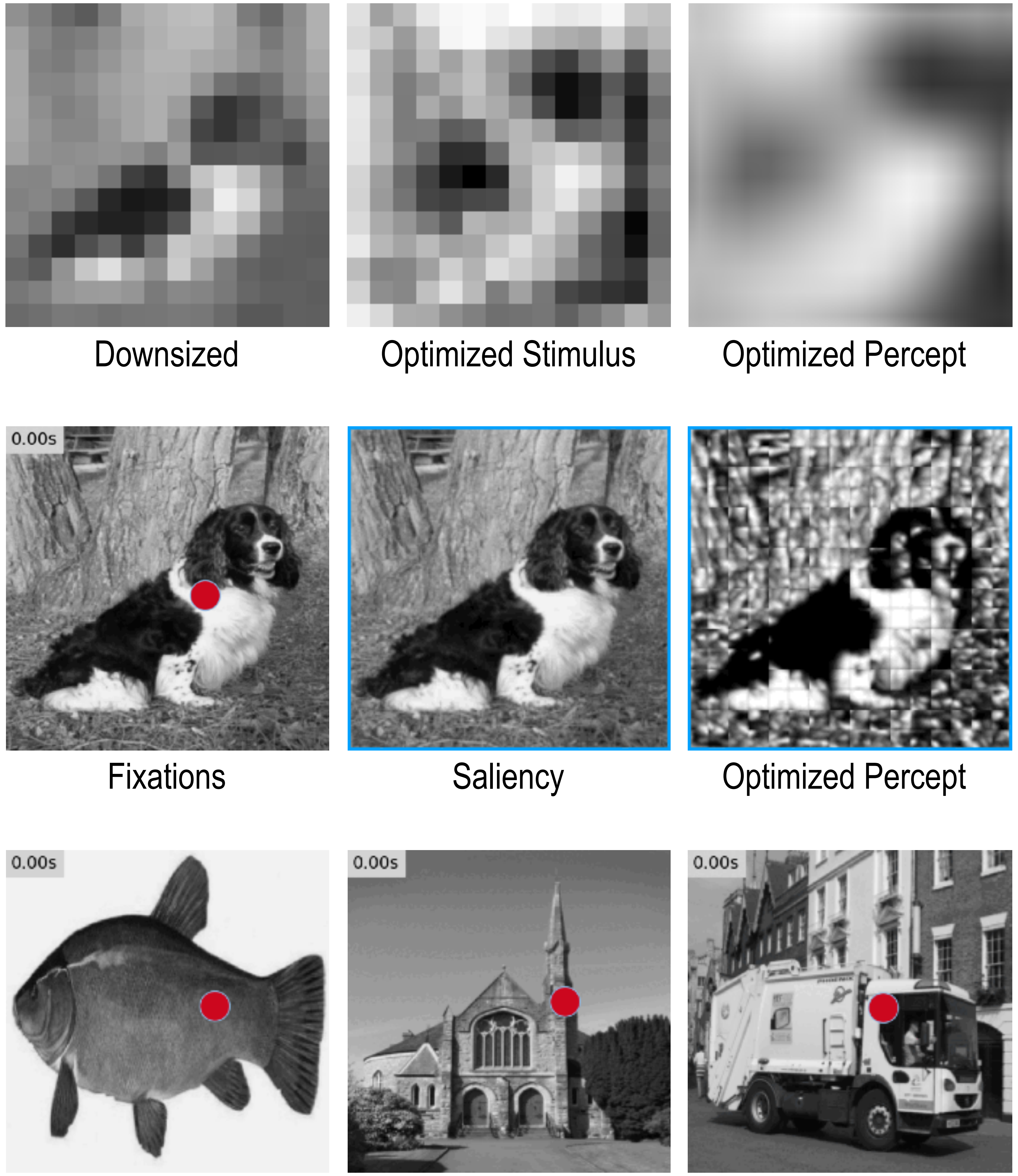
[Wu et al., 2025a]

Visual Fixation-based Simulation

Fixation	Encoder	Linear probing	Accuracy (in %) Param. A	Accuracy (in %) Param. B
✗	✗	Learnable	47.46	38.70
✗	U-Net	Learnable	51.49	40.59
✓	✗	Learnable	85.20	81.99
✓	Linear	Learnable	21.12	21.43
✓	U-Net*	Learnable	53.81	68.79
✓	U-Net	Frozen	85.94	82.62
✓	U-Net	Learnable	**90.85**	**87.72**
Healthy upper bound as comparate:			92.76	







[Wu et al., 2025a]

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